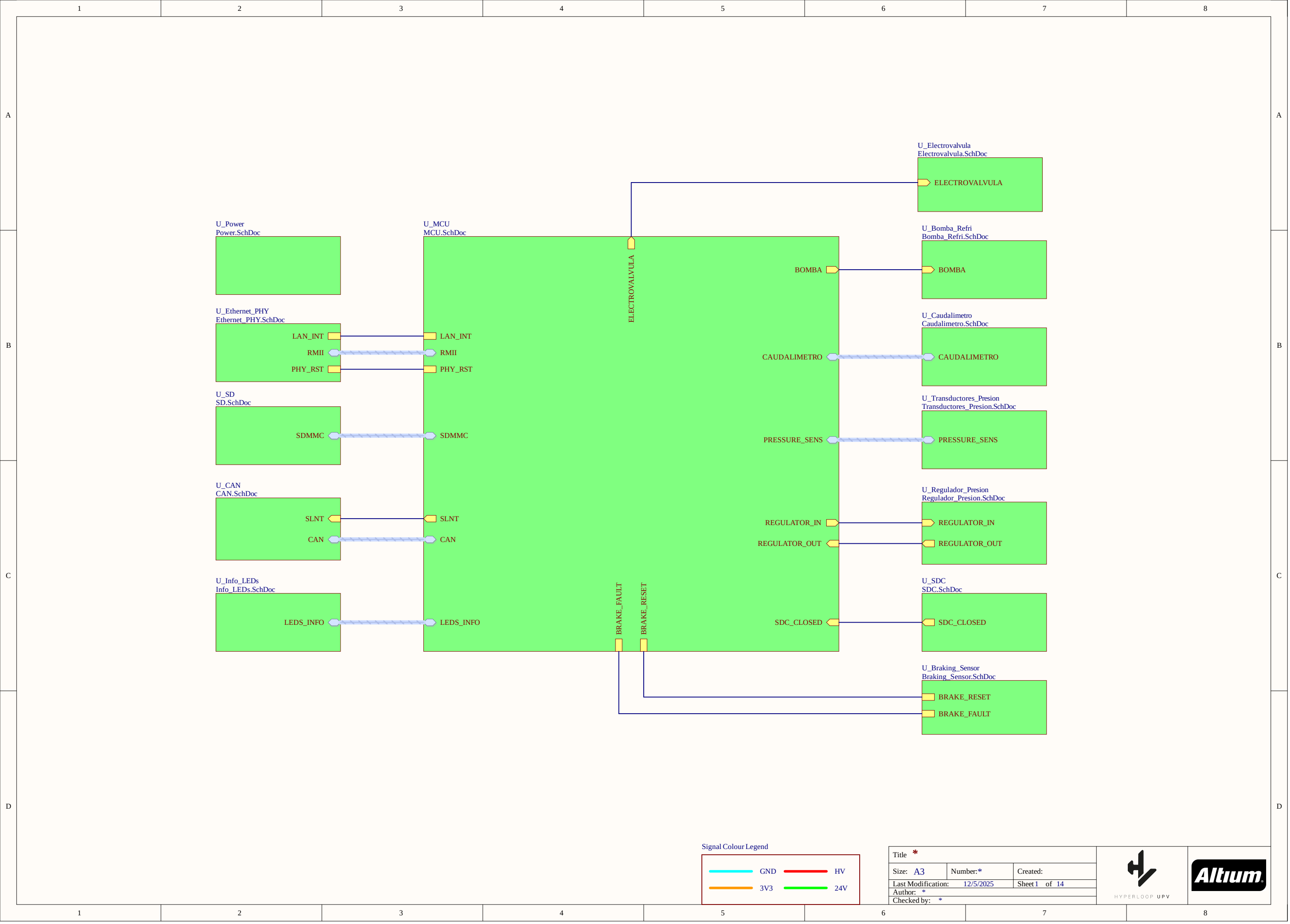
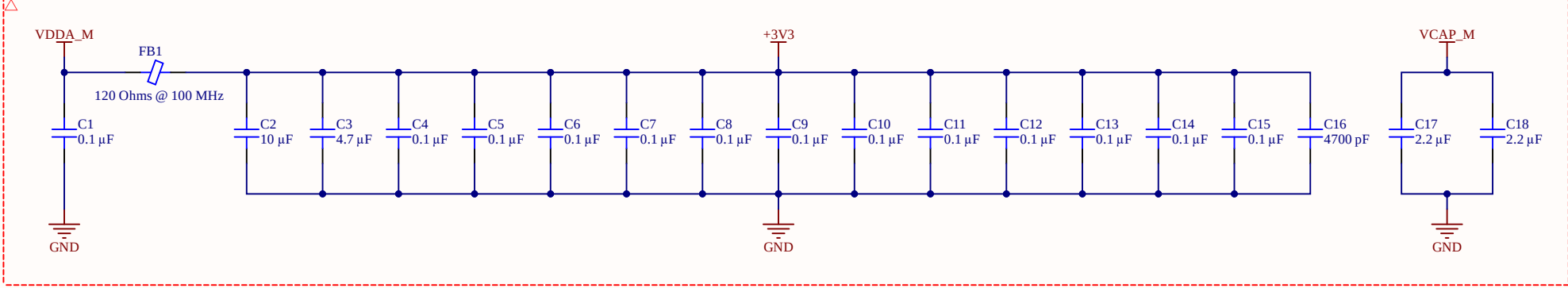
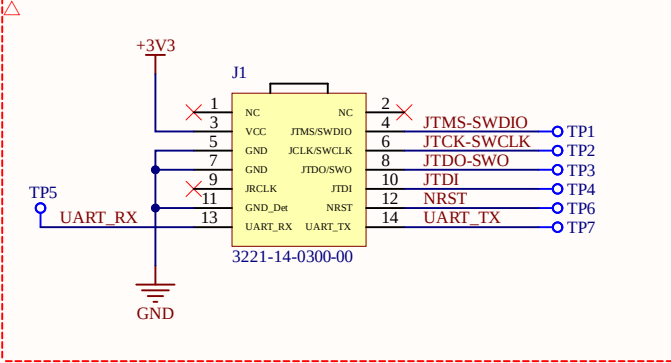




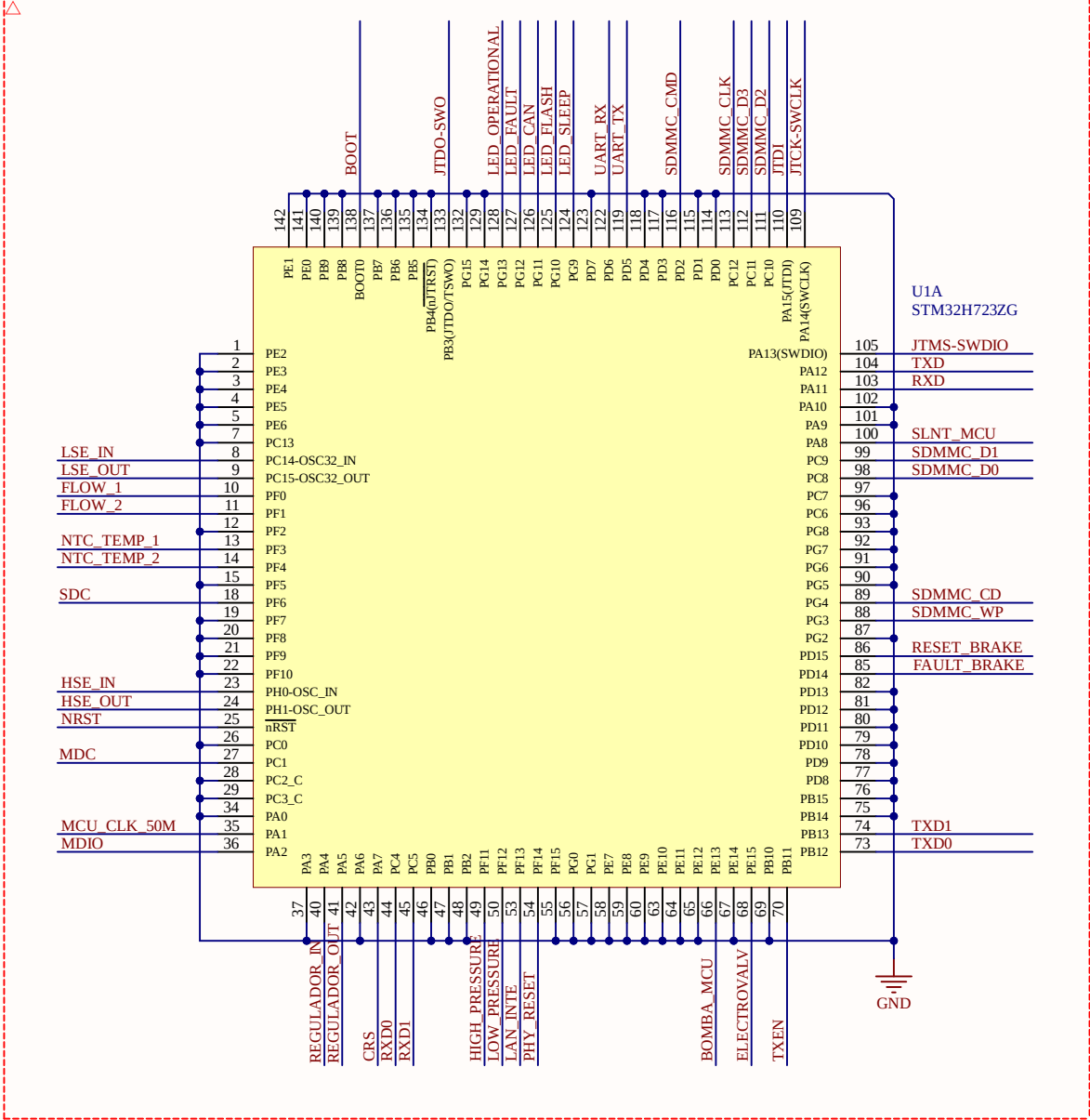
MCU Supply Decoupling



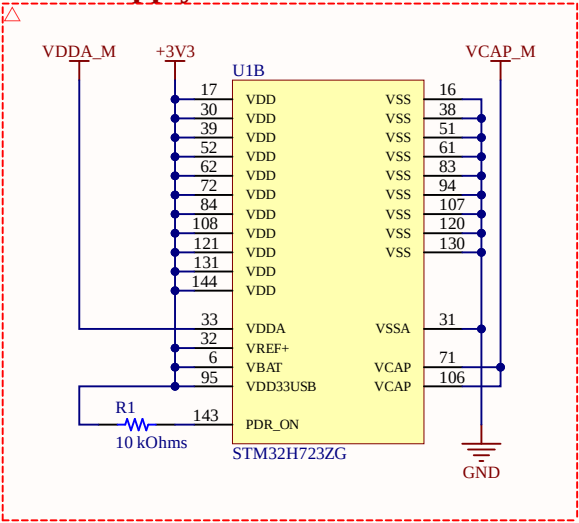
JTAG Interface



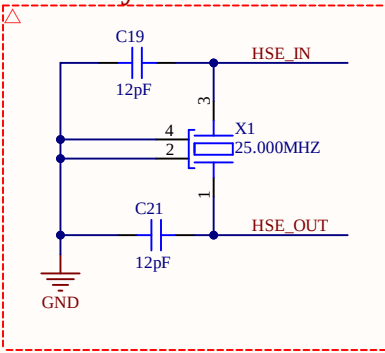
MCU



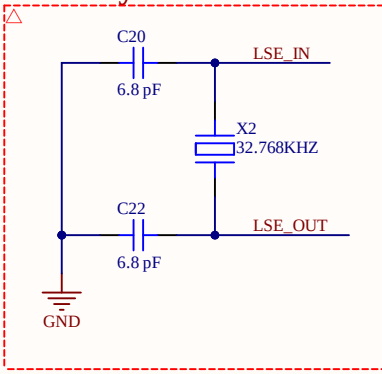
MCU Supply



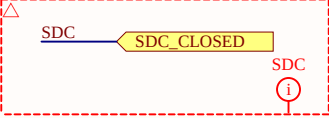
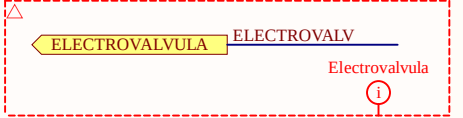
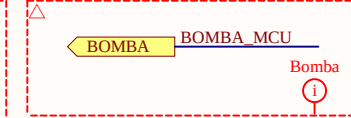
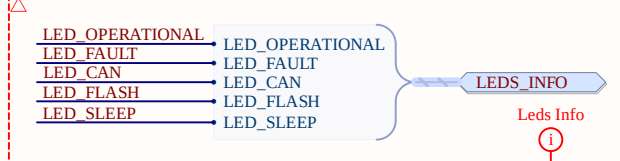
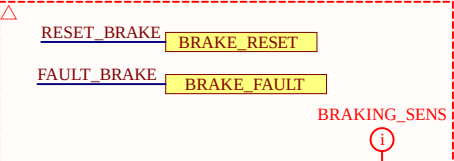
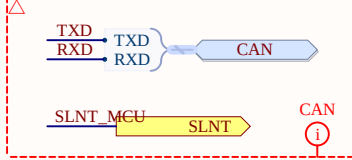
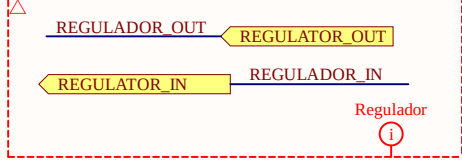
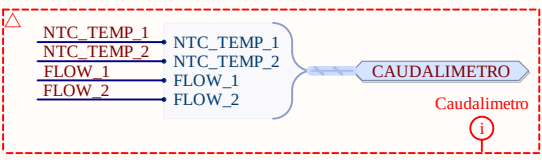
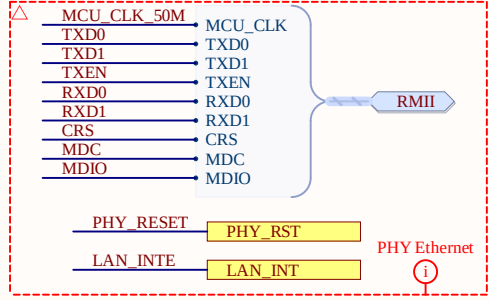
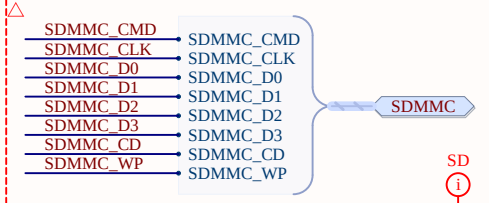
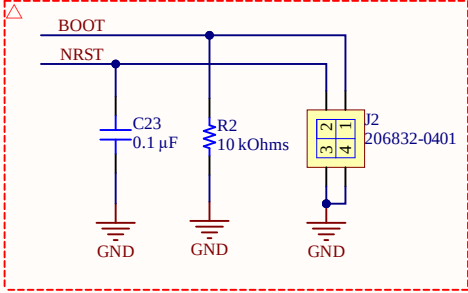
HSE Crystal



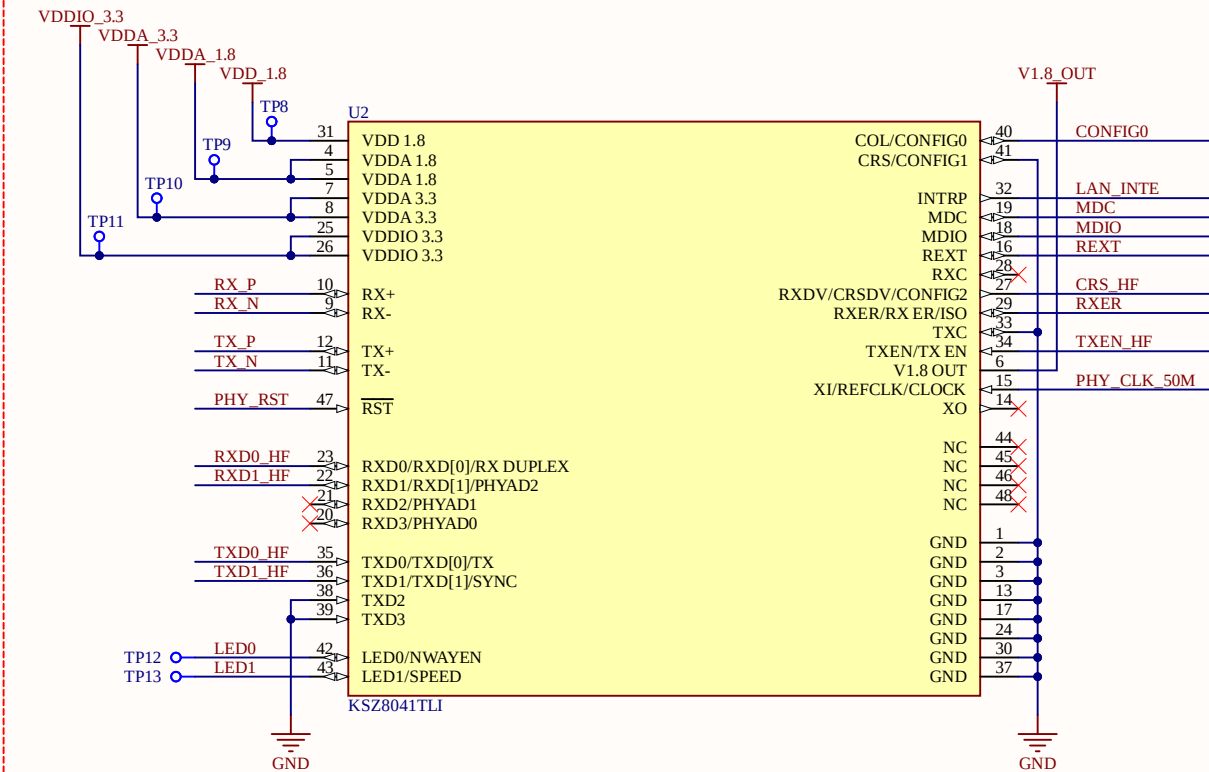
LSE Crystal



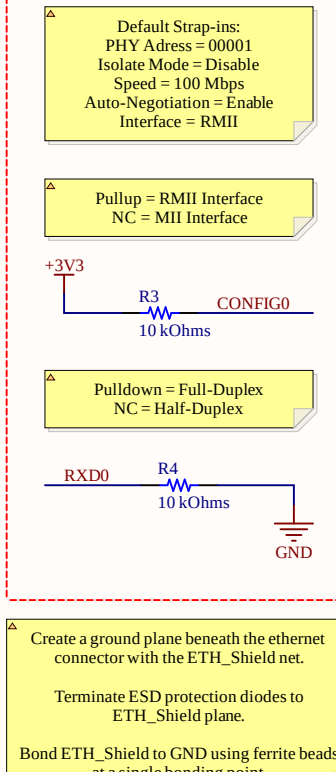
Boot



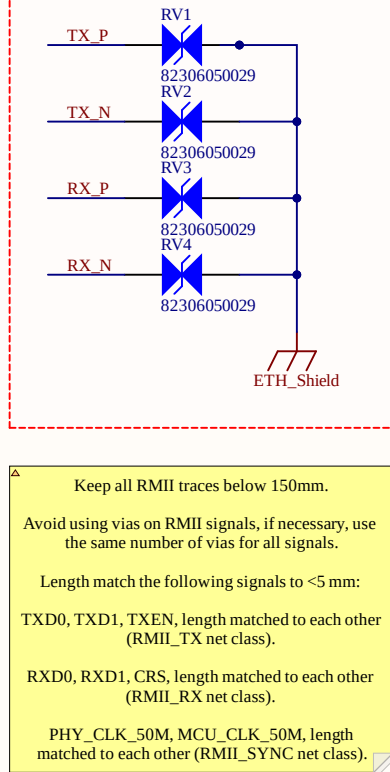
Ethernet Transceiver



Strap-In Options



ESD Protection



Layout Notes

All RMII signals need to be routed using a 50 Ohm coplanar microstrip or stripline impedance profile:

The calculations for trace width have been made according to the manufacturing specs listed in the main architecture sheet of this PCB and assuming a coplanar routing approach (0.1mm clearance to GND).

Note that the signal layers of this PCB are Layer 1, 4 and 6 and the reference layers are Layer 2 and 5. If this is changed, the trace width has to be recalculated.

The results of the JLCPCB online impedance calculator are the following:

Layer 1 trace width = 0.1199 mm
Layer 4 trace width = 0.1476 mm
Layer 5 trace width = 0.1090 mm

The impedance profiles will only be accurate if the following aspects are specified when ordering:

Layers = 6
Inner copper weight = 0.5oz
Outer copper weight = 1oz
PCB thickness = 1.6mm
Manufacturer = JLCPCB
Stack-up Code = JLC0616H-3313A

Route analog RX (RX_P, RX_N) and TX (TX_P, TX_N) pairs as differential signals with a 100 Ohm impedance profile, in a coplanar differential microstrip or stripline configuration.

The calculations for trace width have been made according to the manufacturing specs listed in the main architecture sheet of this PCB and assuming a coplanar differential routing approach (0.1mm clearance to GND, 0.2mm distance between differential signal traces).

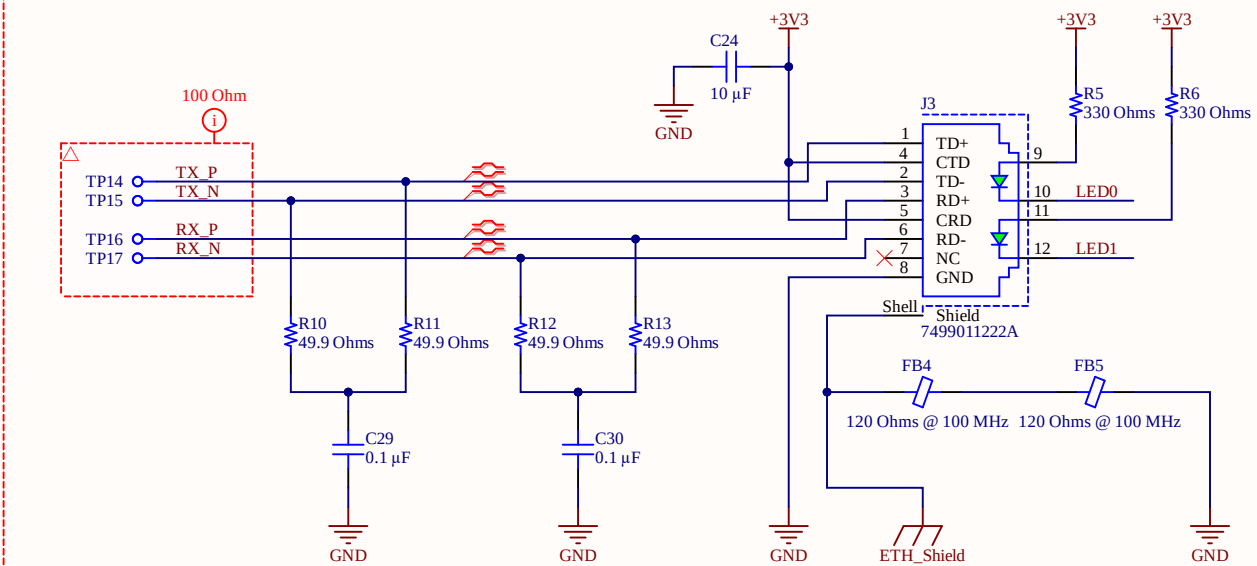
The results of the JLCPCB online impedance calculator are the following:

Layer 1 trace width = 0.1090 mm
Layer 4 trace width = 0.1476 mm
Layer 5 trace width = 0.1090 mm

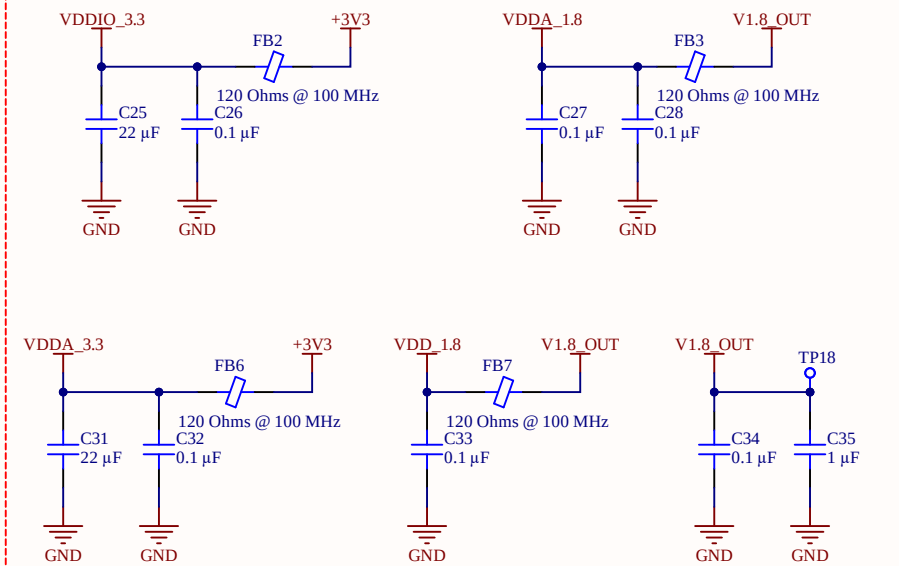
The impedance profiles will only be accurate if the following aspects are specified when ordering:

Layers = 6
Inner copper weight = 0.5oz
Outer copper weight = 1oz
PCB thickness = 1.6mm
Manufacturer = JLCPCB
Stack-up Code = JLC0616H-3313A

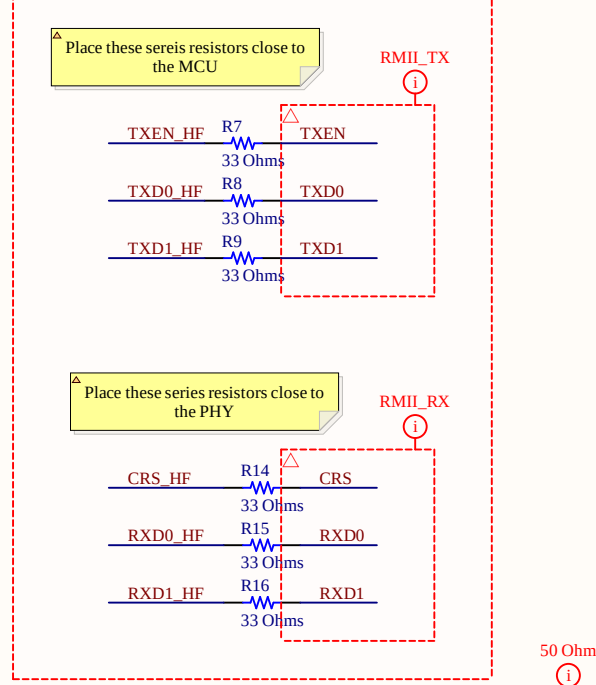
Analog Interface



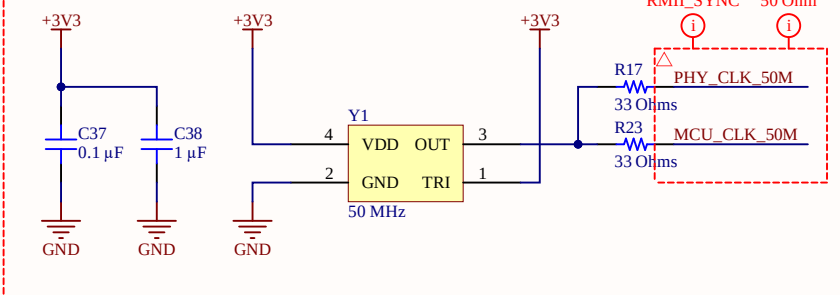
Transceiver Supply Decoupling



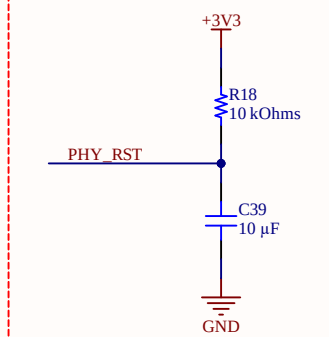
Slew Rate Control



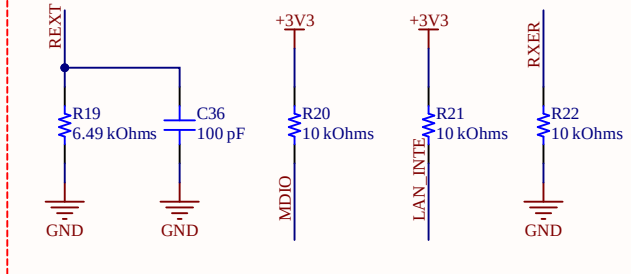
50 MHz Clock



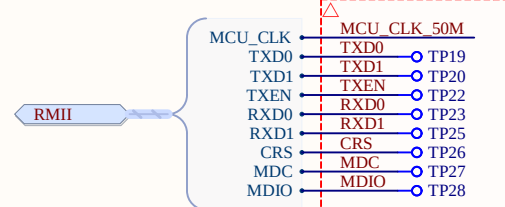
Reset



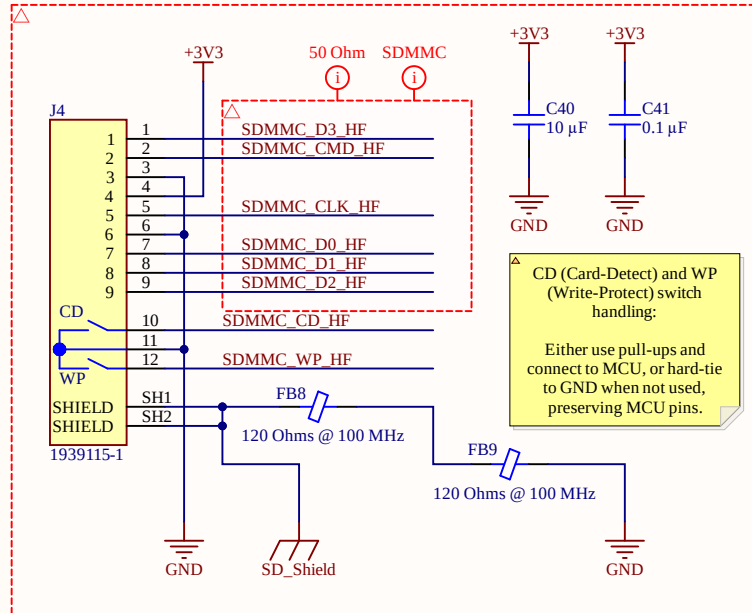
Biasing



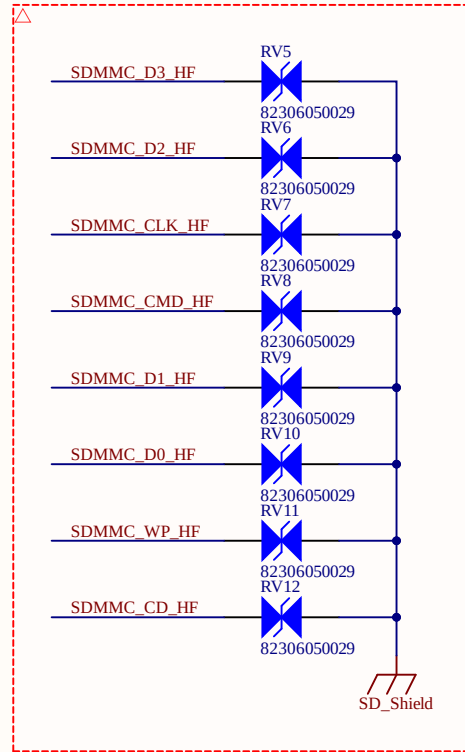
LAN_INT LAN_INTE TP21
PHY_RST PHY_RST TP24



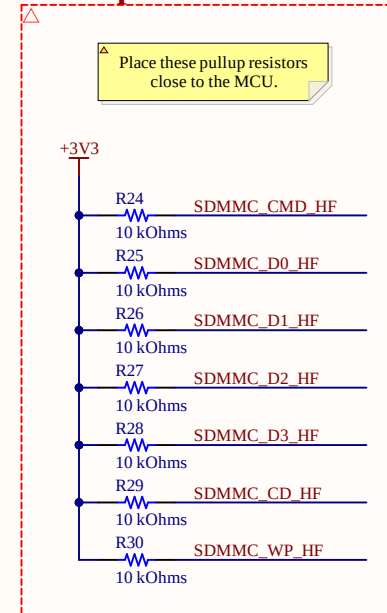
SD Card Slot



ESD Protection



Pull-Up



Layout Notes

- Keep SDMMC traces below 120mm.
- Avoid using vias on SDMMC signals, if necessary, use the same number of vias for all signals.
- Length match the following signals to <10 mm:
 - CMD, CLK, D0, D1, D2, D3, length matched to each other (SDMMC net class).
- Create a ground plane beneath the SD card connector with the SD_Shield net.
- Terminate ESD protection diodes to SD_Shield plane.
- Bond SD_Shield to GND using ferrite beads at a single bonding point.

△ The following signals need to be routed using a 50 Ohm coplanar microstrip or stripline impedance profile:

The calculations for trace width have been made according to the manufacturing specs listed in the main architecture sheet of this PCB and assuming a coplanar routing approach (0.1mm clearance to GND).

Note that the signal layers of this PCB are Layer 1, 4 and 6 and the reference layers are Layer 2 and 5. If this is changed, the trace width has to be recalculated.

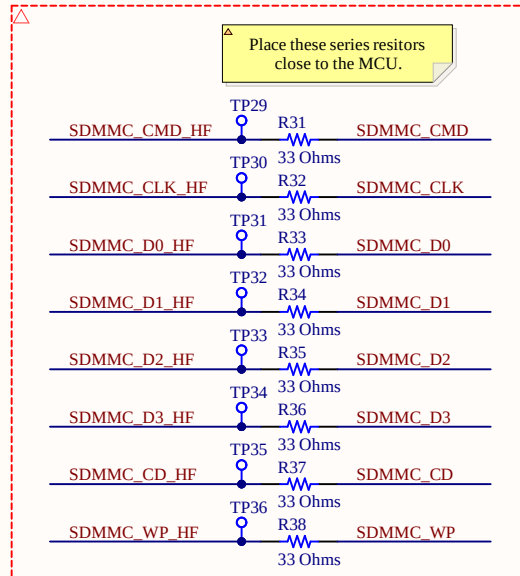
The results of the JLCPCB online impedance calculator are the following:

Layer 1 trace width = 0.1199 mm
Layer 4 trace width = 0.2149 mm
Layer 6 trace width = 0.1199 mm

The impedance profiles will only be accurate if the following aspects are specified when ordering:

Layers = 6
Inner copper weight = 0.5oz
Outer copper weight = 1oz
PCB thickness = 1.6mm
Manufacturer = JLCPCB
Stack-up Code = JLC0616H-3313A

Slew Rate Control

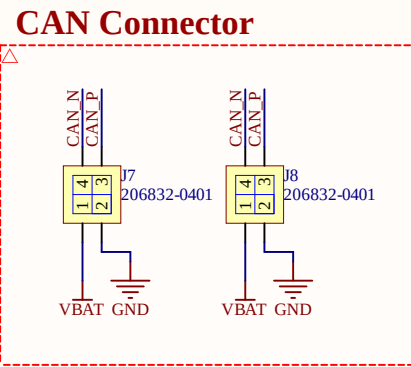
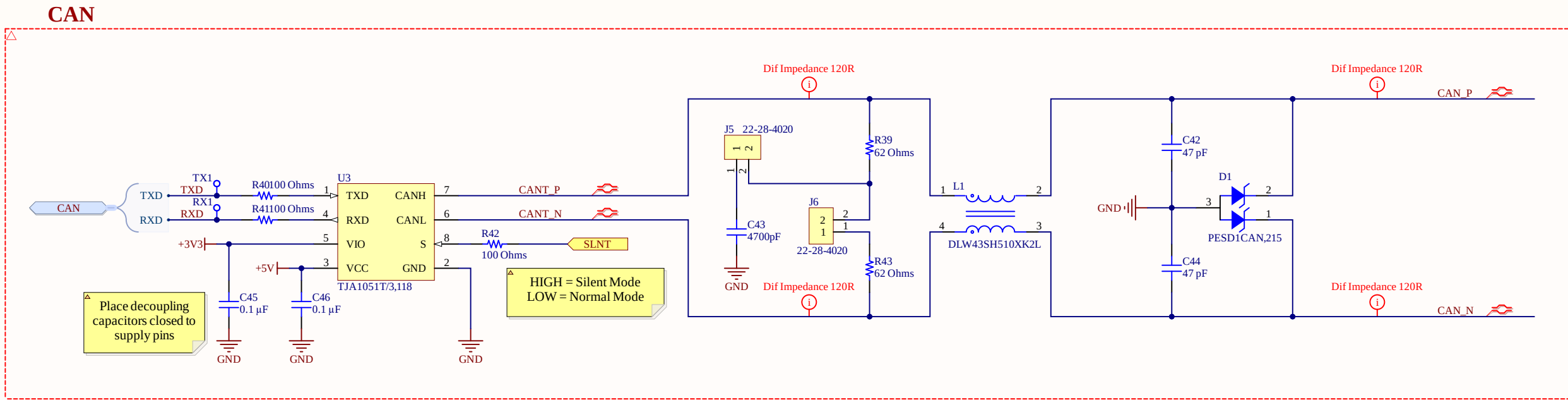


SDMMC

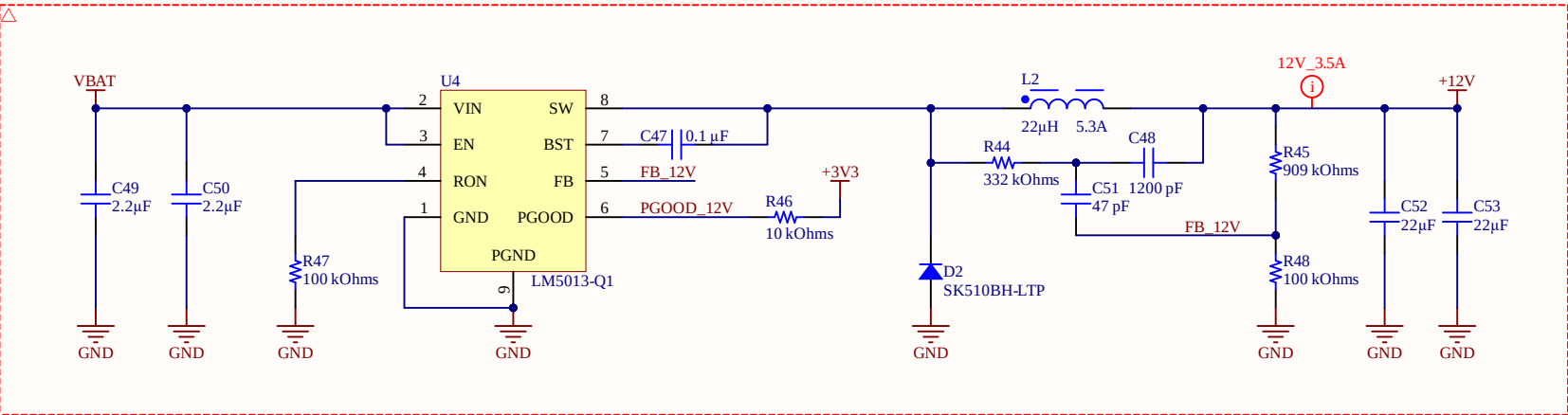
SDMMC_CMD	SDMMC_CMD
SDMMC_CLK	SDMMC_CLK
SDMMC_D0	SDMMC_D0
SDMMC_D1	SDMMC_D1
SDMMC_D2	SDMMC_D2
SDMMC_D3	SDMMC_D3
SDMMC_CD	SDMMC_CD
SDMMC_WP	SDMMC_WP

Title *		
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Last Modification: 12/5/2025	Sheet4 of 14	
Author: *		
Checked by: *		

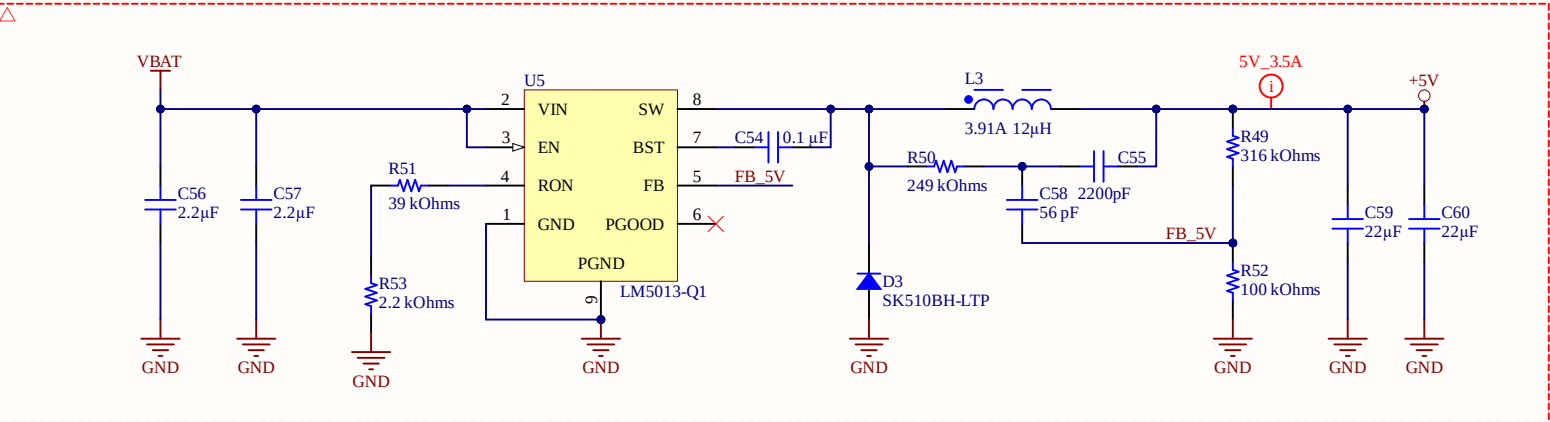




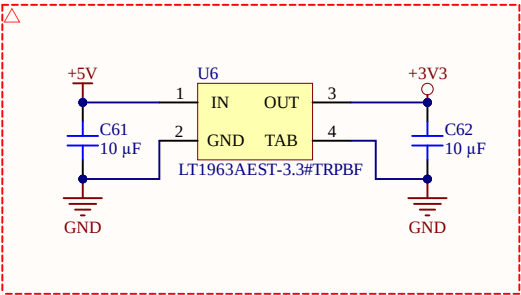
+12V Power Supply



+5V Power Supply



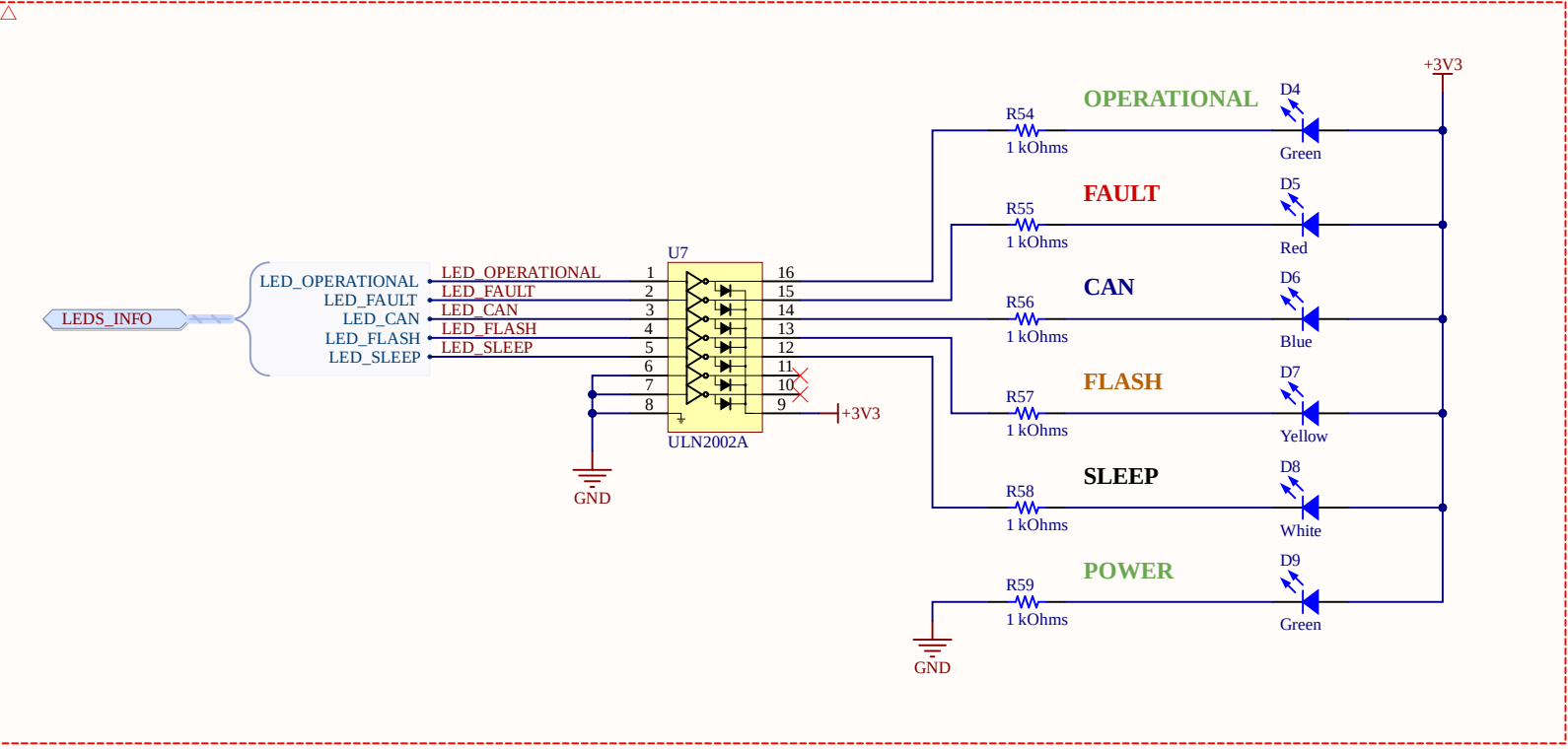
+3V3 Power Supply



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Size: A3	Number:*	Created:
Last Modification: 12/5/2025	Sheet6 of 14	
Author: *		
Checked by: *		



Information LEDs



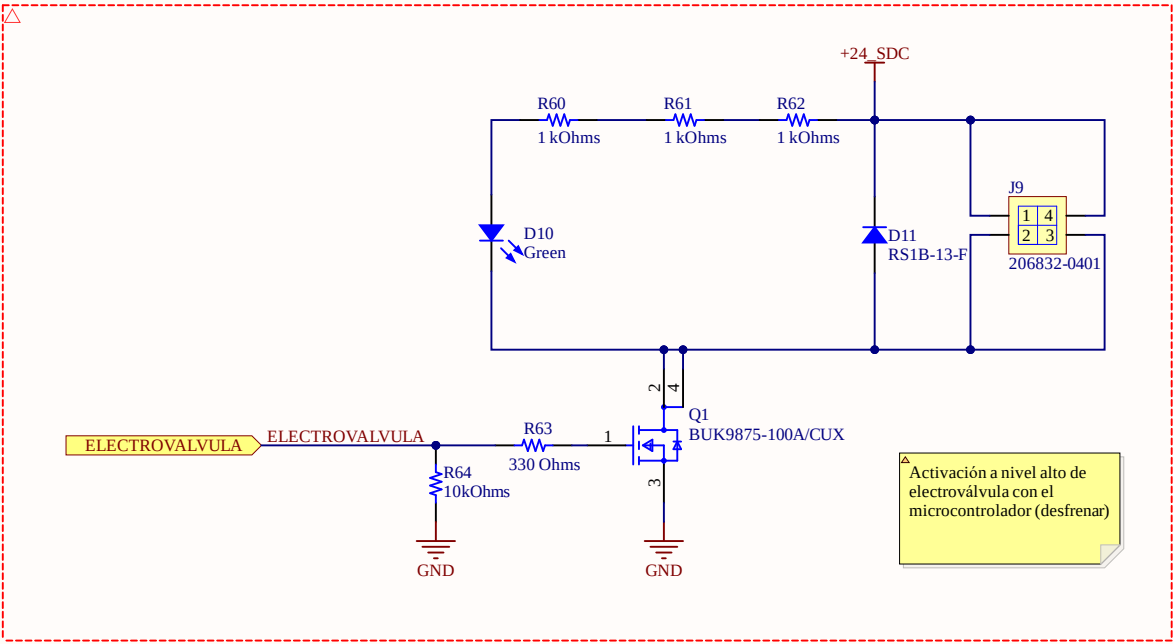
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Last Modification: 12/5/2025	Sheet 7 of 14	
Author: *		
Checked by: *		



HYPERLOOP UPV



ELECTROVALVULA

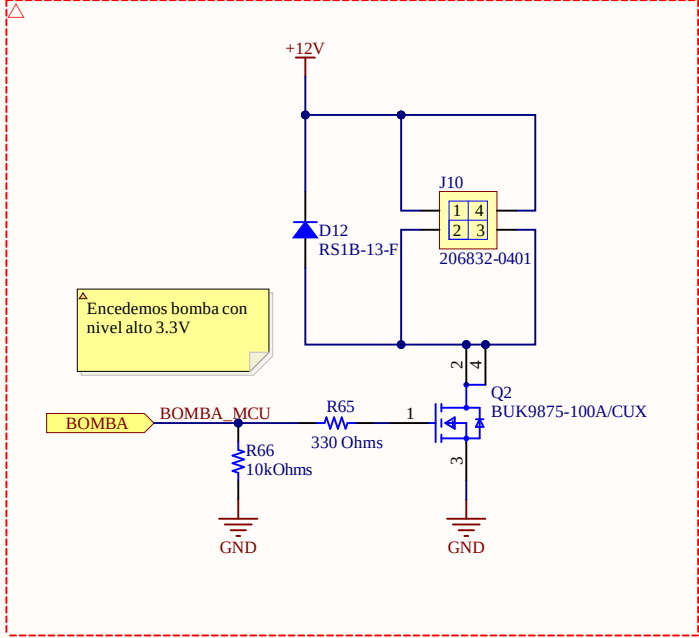


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Last Modification: 12/5/2025	Sheet8 of 14		
Author: *			
Checked by: *			


HYPERLOOP UPV



BOMBA REFRIGERACION

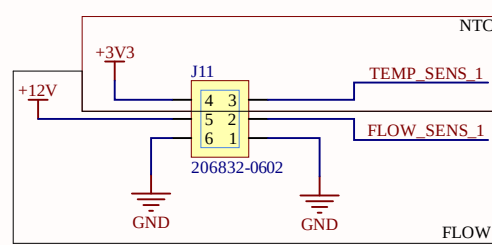


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Size: A4	Number:*	Created:
Last Modification: 12/5/2025	Sheet9 of 14	
Author: *		
Checked by: *		



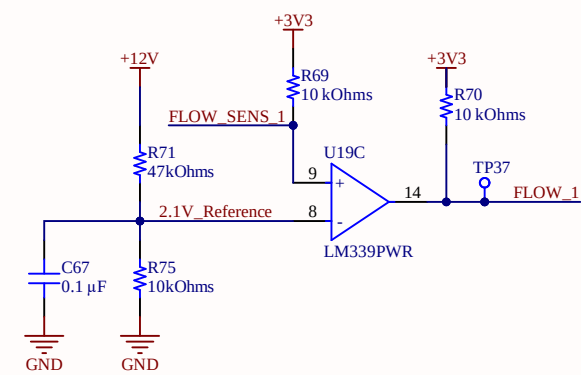
Caudalimetro_1

Conector_1

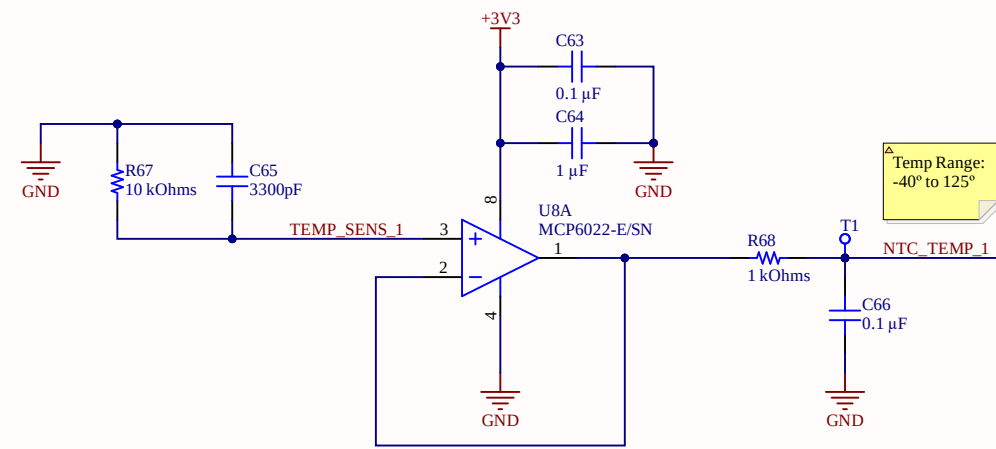


Como no sabemos si la salida digital del sensor de caudal es open drain o push pull, no nos podemos arriesgar a dañar el micro, por eso usamos este comparador, la idea es cuando se sepa en caso de que sea push pull no soldaremos la pull-up que se encuentra en el pin 5 (entrada positiva) y en caso de ser open drain soldaremos el comparador junto a la pull-up

Sensor_Caudal_1



Sensor_NTC_1

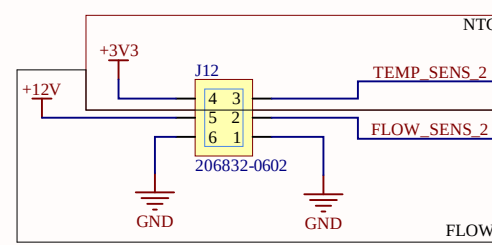


Adjunto aqui el link de la pagina web, en ella encontramos el datasheet y el manual: <https://shop.alphacool.com/en/shop/controllers-and-sensors/flow-sensor/17558-alphacool-es-flow-and-temperature-indicator-highflow-with-rpm-signal>

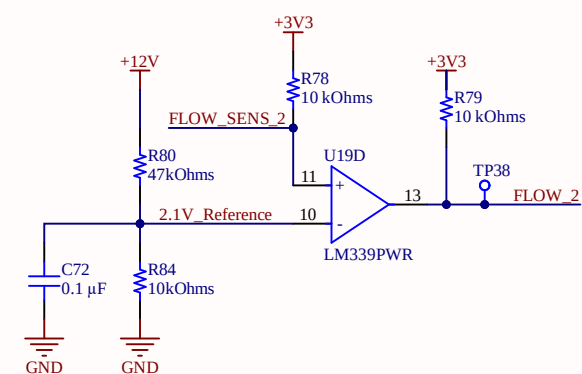
Explicando un poco este circuito del caudalimetro tenemos por un lado, un sensor de temp, el cual es una NTC que lee la temperatura del agua de la refrigeracion, y por otro lado tenemos un sensor de caudal, el cual es un sensor Hall con Tachometer/RPM Signal, esto indica que es salida digital Open Drain (por eso la pull-up), ademas este salida del sensor la deberemos de conector a un INPUT CAPTURE de la STM32, debido a que el codigo de la STM32 no mediremos voltaje, si no que el micro contará los pulsos que le llegan en periodo determinado para calcular la frecuencia (RPM)

Caudalimetro_2

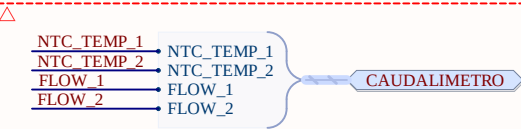
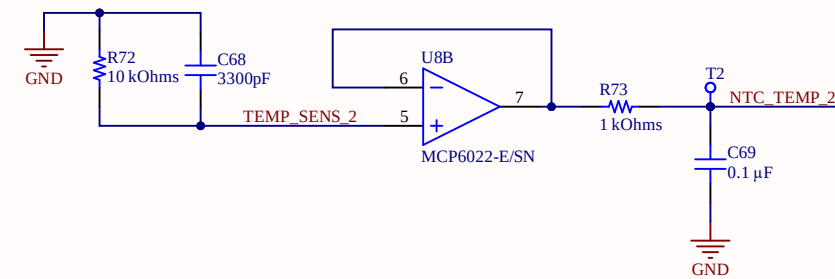
Conector_2



Sensor_Caudal_2

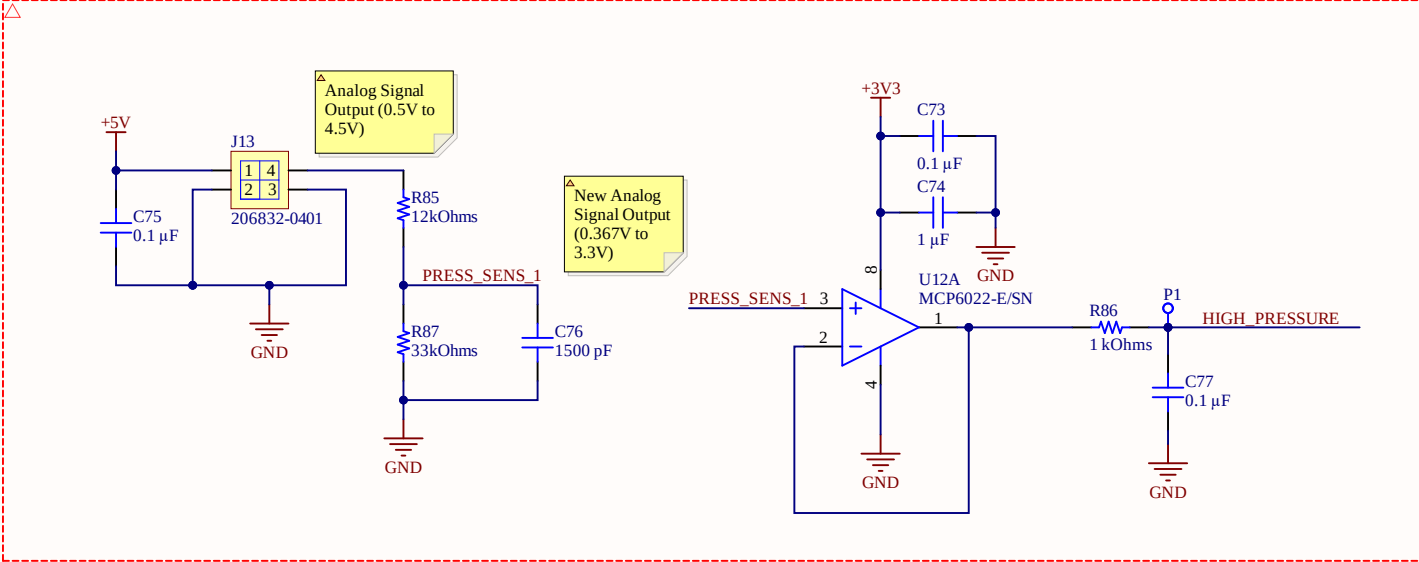


Sensor_NTC_2



Title		
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Date:	5/11/2026	Sheet of
File:	Caudalimetro.SchDoc	Drawn By:

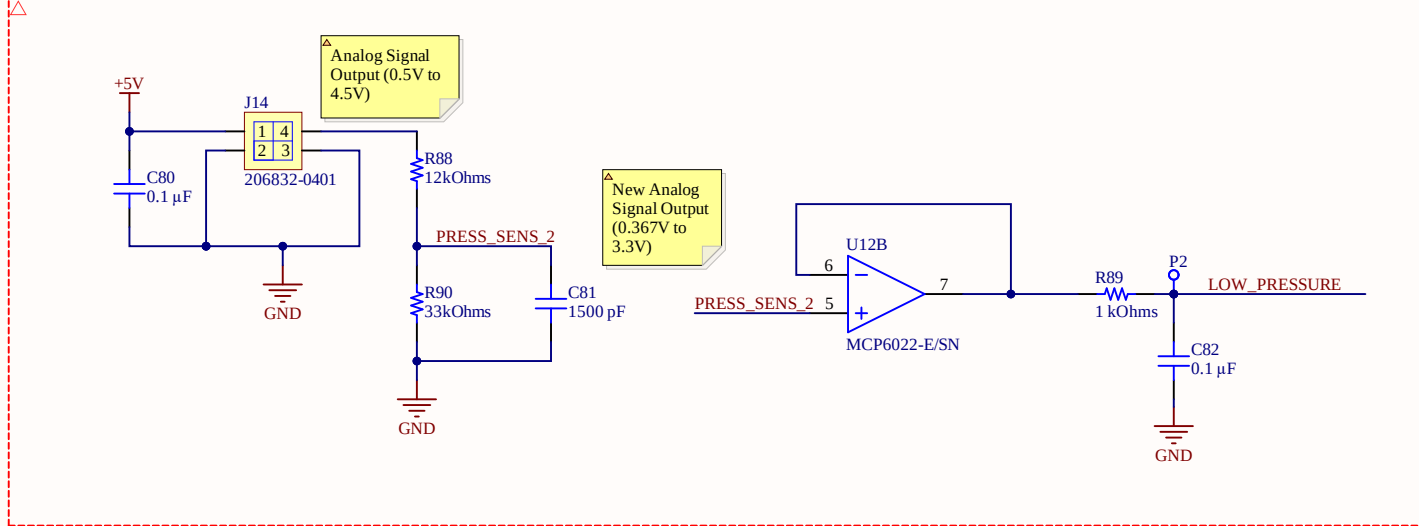
Transductor_Alta_Presion



Pin Out: Red = Vdc (5V) 10% to 90% Vs
Black = GND
Green =No se usa
White = Analog Signal Output (0.5V to 4.5V)

Ratiometric
Rango de presion (0-5000 Psi)

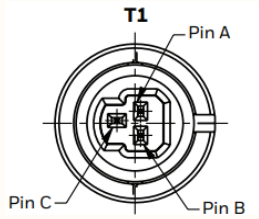
Transductor_Baja_Presion



Pin Out:
Pin A = Vdc (5V) 10% to 90% Vs
Pin B = GND
Pin C = Analog Signal Output (0.5V to 4.5V)

Ratiometric
Rango de presion (0-10 Bar)

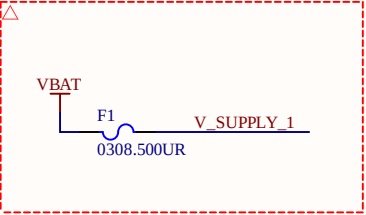
Como es Sealed Gage (parte de 1 atm = 1.013 Bar) no parte realmente de 0 a 10 Bar, si no que el rango de presion real (1.013-11.013 Bar)



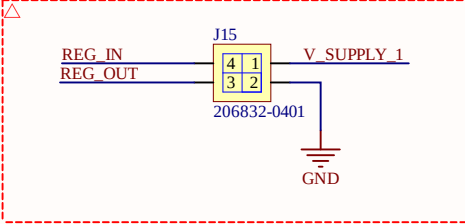
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Size: A4	Number:*	Created:
Last Modification: 12/5/2025	Sheet 11 of 14	
Author: *		
Checked by: *		



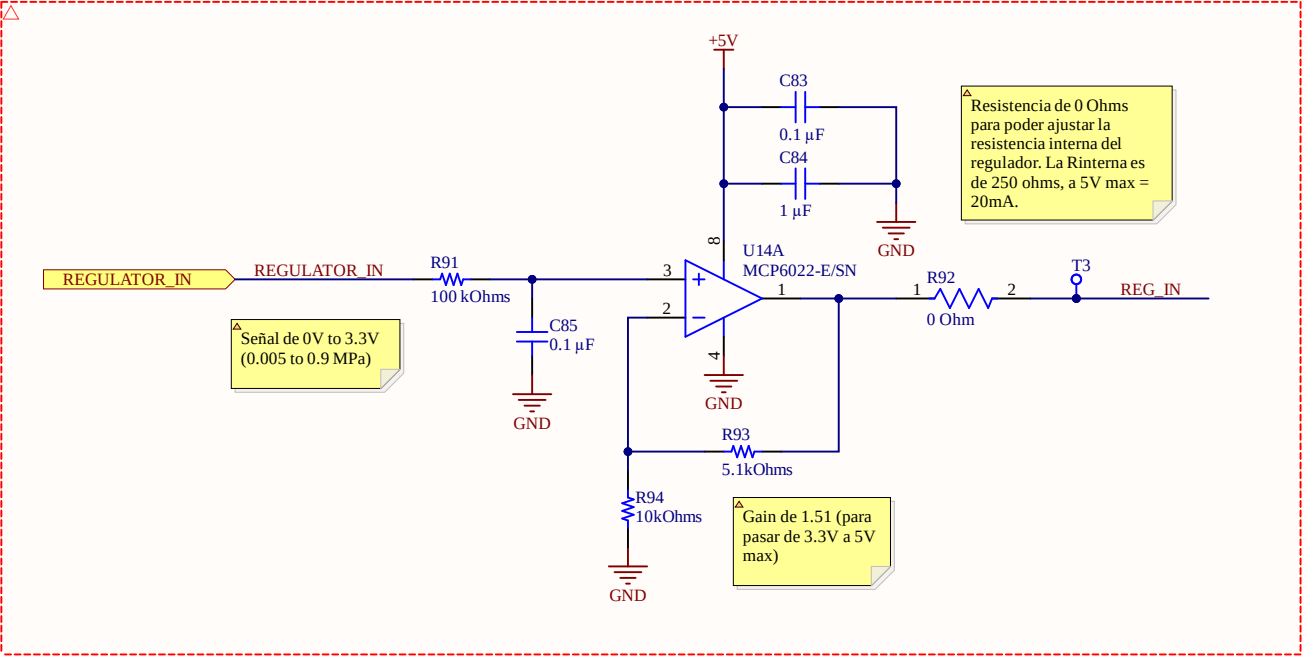
Supply



Conector



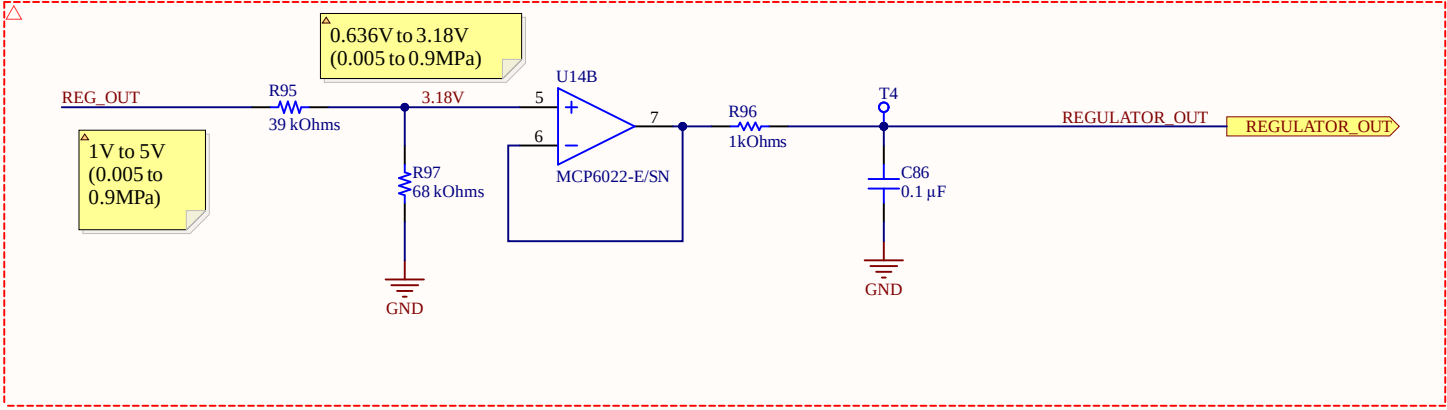
Regulator IN



Pin Out:
Brown = Vdc (24V)
Blue = GND
White = Analog Signal IN, sink type (4mA to 20mA)
Black = Analog Signal Monitoring (1V to 5V)

Rango de presion (0.005 to 0.9 MPa)

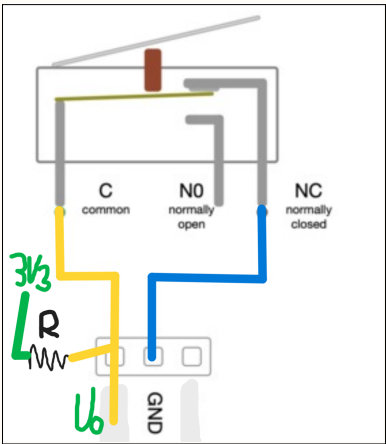
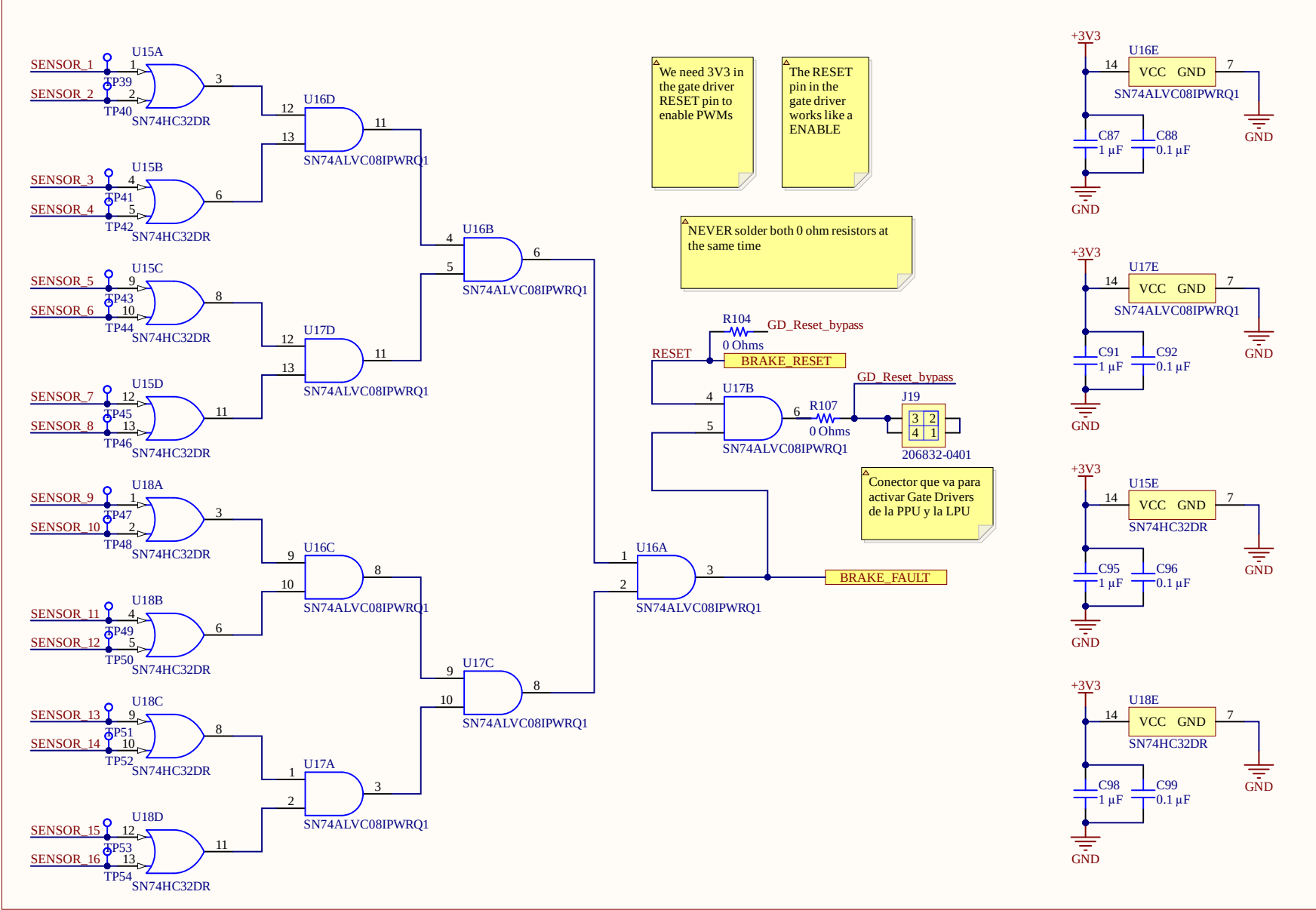
Regulator OUT



Title *		
Size: A4	Number:*	Created:
Last Modification: 12/5/2025	Sheet 12 of 14	
Author: *		
Checked by: *		



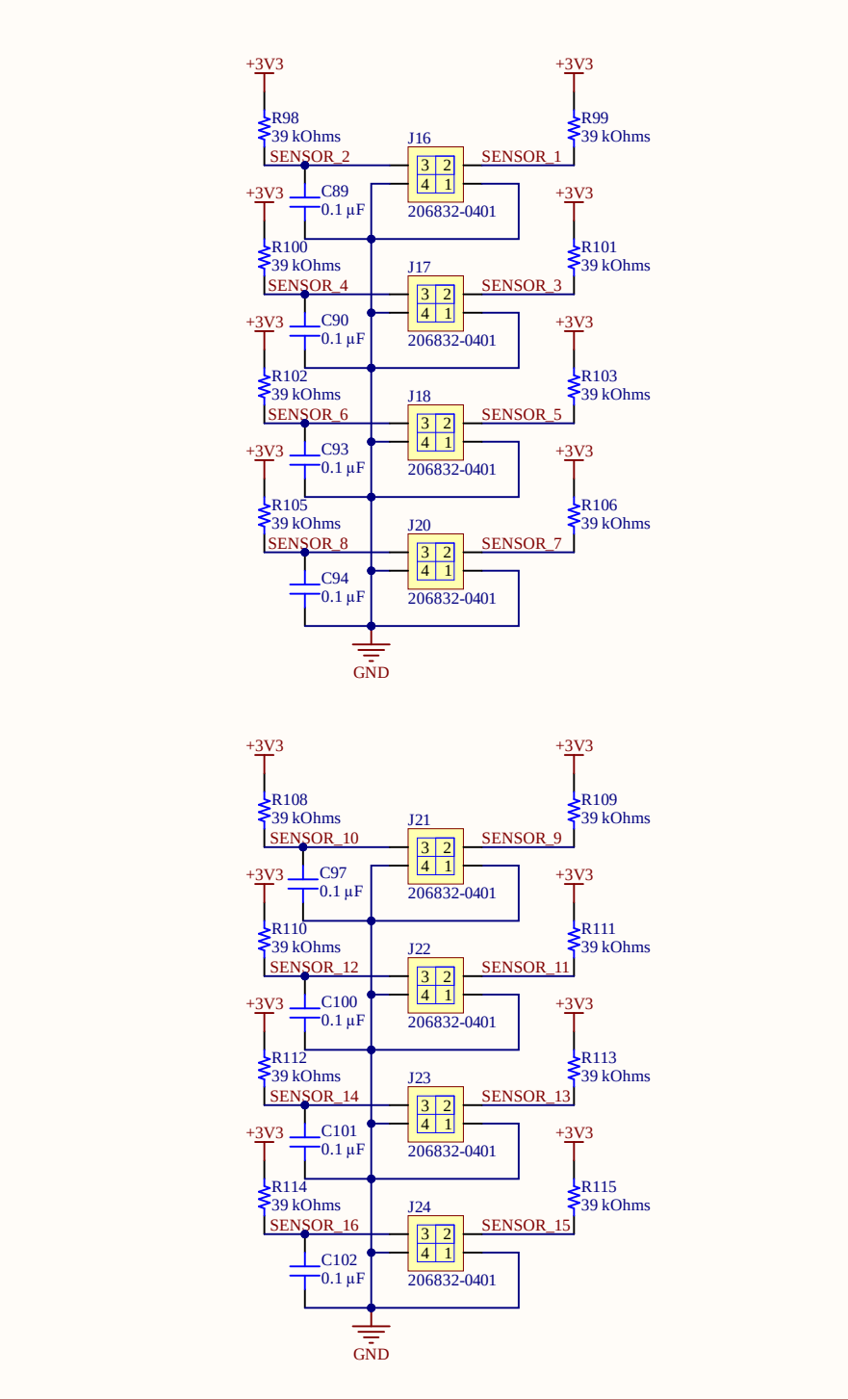
Limit switch logic



We should connect the NC pin of the limit switch sensor to GND and the NO pin floating

When the vehicle is braked, the limit switch is NOT pressed, it is pressed when the brakes are released

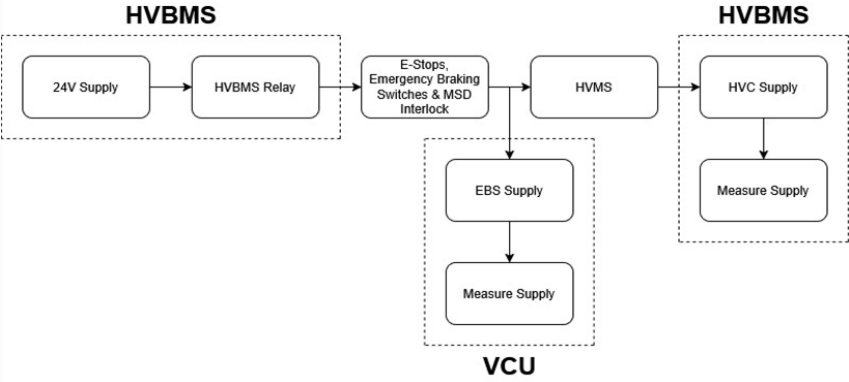
Limit switch connector



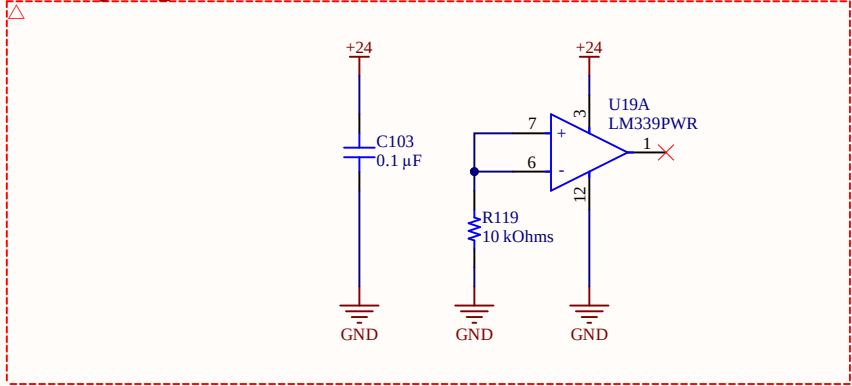
Title *		
Size: A3	Number:*	Created:
Last Modification: 12/5/2025	Sheet13 of 14	
Author: *		
Checked by: *		



Shut-down Circuit (SDC)



Decoupling & Unused Channels



SDC Input & Measurement

